

NLP-2024Z - PROJECT

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This project focuses on analyzing a dataset of Polish legal clauses to build a machine learning classifier that can automatically identify abusive (unfair) clauses in contracts. The data I was provided comes from the HuggingFace database: <https://huggingface.co/datasets/laugustyniak/abusive-clauses-pl/viewer/abusive-clauses-pl/validation>.

The goal is to create a system that achieves strong performance, particularly in F1-score, to balance precision (avoiding false accusations) and recall (catching all potential abuses).

My code can be found in this link:

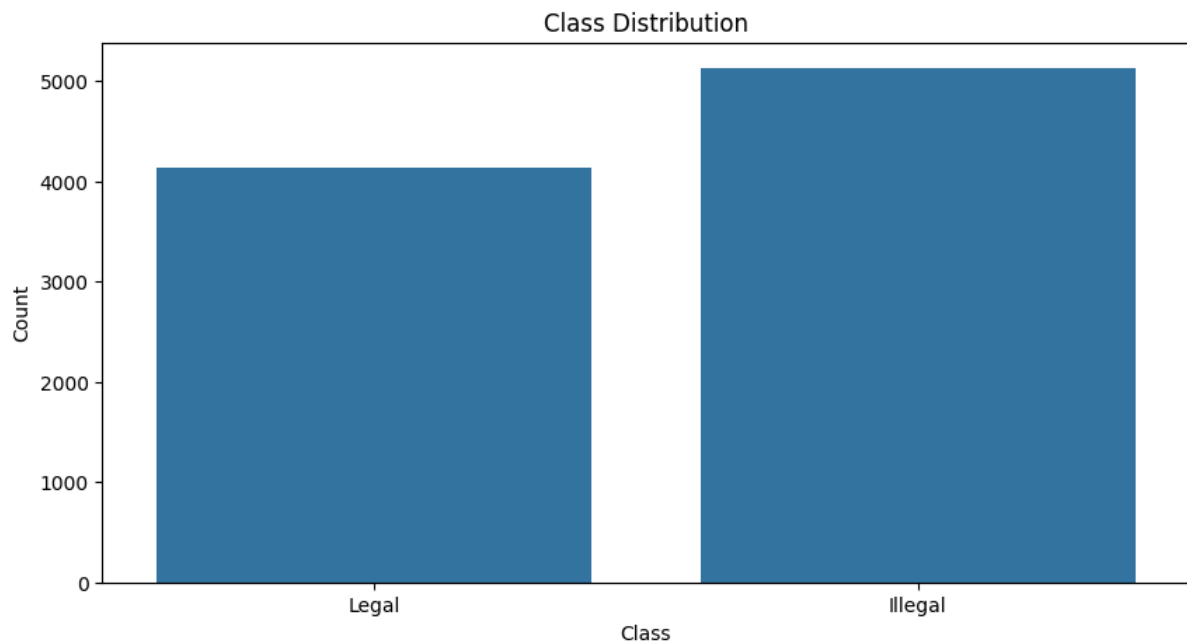
https://github.com/RusieckiFilip/Inzynieria_Obliczeniowa_UW/blob/main/NLP_OBL_I/NZ.ipynb

I started by analyzing a dataset, which contains legal texts labeled as either "Legal" or "Illegal." My first challenge was the imbalance in the data. Most clauses were labeled as "Legal," which could bias the model. To address this, I experimented with techniques like adjusting class weights during training and making the validation set more balanced like in the training set.

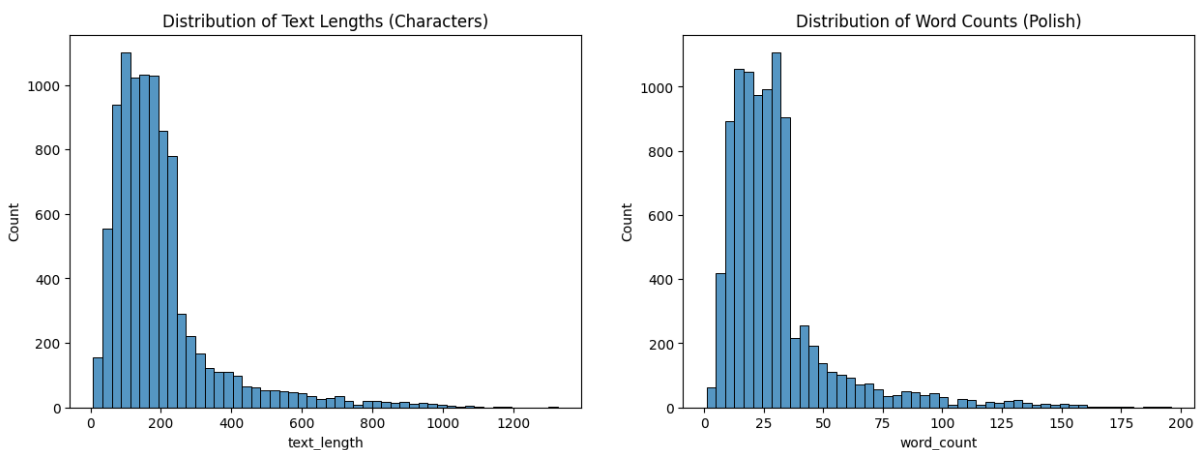
```
TRAIN Set (n=4284):  
Illegal clauses (0): 1946 (45.42%)  
Legal clauses (1): 2338 (54.58%)  
  
VAL Set (n=1519):  
Illegal clauses (0): 1063 (69.98%)  
Legal clauses (1): 456 (30.02%)  
  
TEST Set (n=3453):  
Illegal clauses (0): 1120 (32.44%)  
Legal clauses (1): 2333 (67.56%)  
  
Class weights for imbalance: [2.2, 1.83]
```

I calculated weights for this imbalance in the training set, so I could use it later in creating a model.

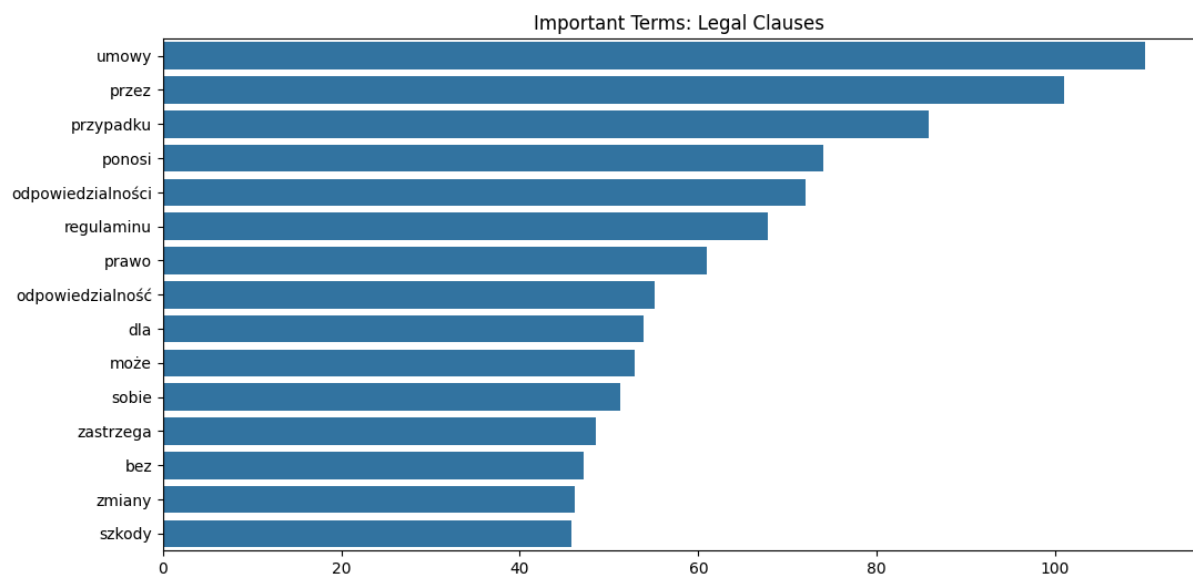
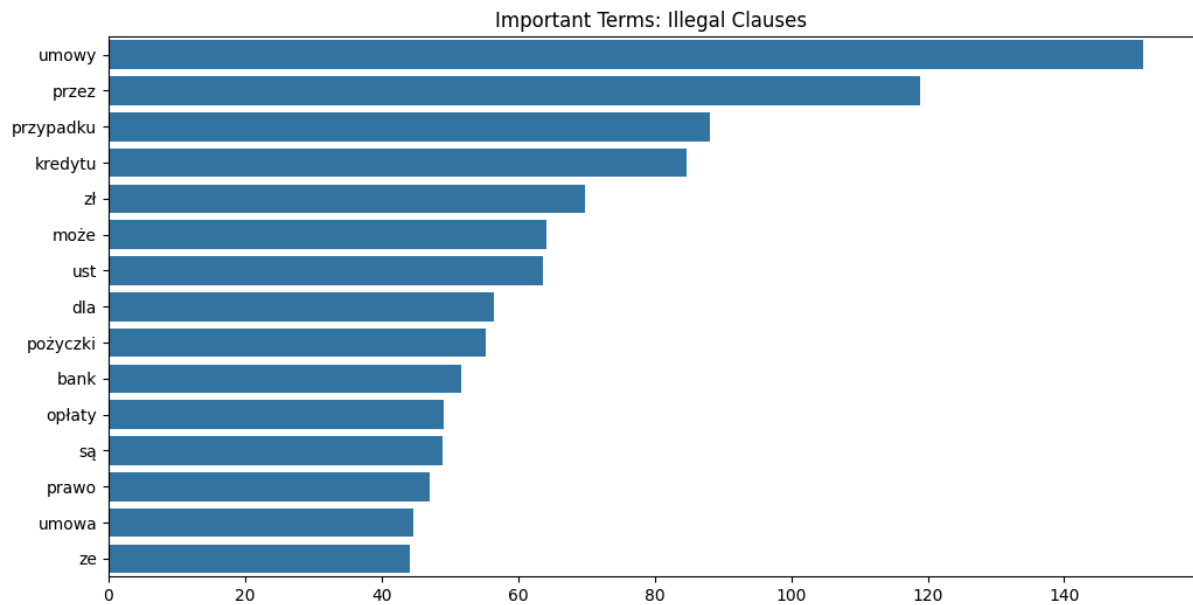
The Class distribution for whole dataset (training+validation+test) is very similar to training set (around 55% for legal cases)



Next, I focused on analysing data. Using spaCy's Polish language model, I cleaned the data by removing common stopwords and tokenizing the text into meaningful units.



Most of the data had around 20 to 40 words, or around 100 to 250 characters.



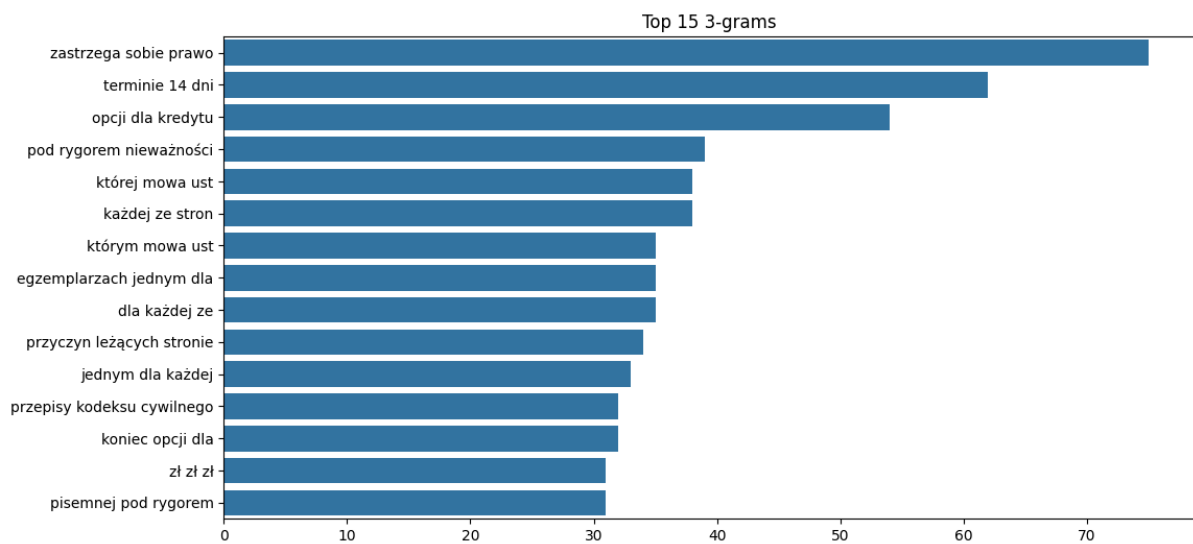
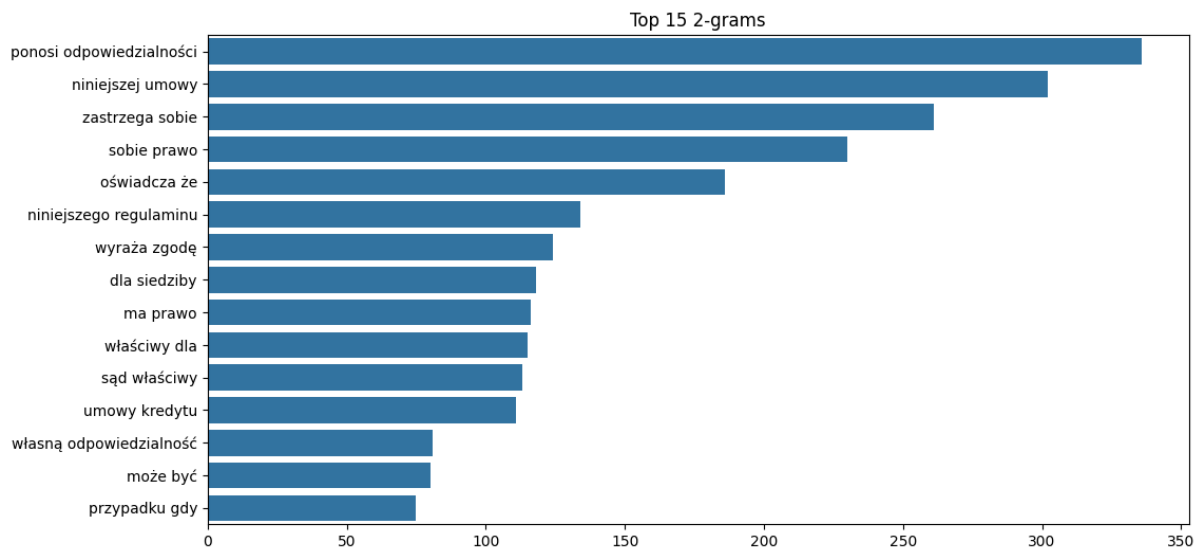
Most of the words are similar in legal and illegal cases, but for legal “zastrzega” (reserves) is more frequent and for illegal we can see that “może” (it can) is far more frequent. It can be the case that abusive clauses are more often written in formalisms than in legal language. It can also be noticed that illegal clauses often include credits or loans.

I also did word clouds for illegal and legal clauses:



No clear conclusions can be drawn from them about the structure or key words in legal and illegal clauses. The only conclusion I could see is that the most repetitive words are the greater part in illegal clauses and in legal ones there is a greater variety of words, therefore words such as 'przypadku' (case) or 'niniejszej umowy' (this agreement) are not as large.

I also showed the top 15 Bigrams and Trigrams that are in these texts.



After analysing data I started preprocessing data to use in the models. I converted the text into numerical features using Term Frequency - Inverse Document Frequency and later fine-tuned 3 BERT models (a Polish RoBERTa model - clarin-pl/roberta-polish-kgr10, BERT fine-tuned for legal cases - nlpaueb/legal-bert-base-uncased, and Base BERT for reference - bert-base-multilingual-cased) to capture deeper contextual relationships. Training the classifier required balancing precision and recall. Since missing abusive clauses (false negatives) or incorrectly flagging fair ones (false positives) both have serious implications, I prioritized optimizing the F1-score. I worked with a balanced validation set, doing weights for class imbalances and doing models without.

```
Original Train (n=4284):  
Illegal (0): 1946 (45.4%)  
Legal (1): 2338 (54.6%)  
  
New Validation (n=1519):  
Illegal (0): 667 (43.9%)  
Legal (1): 852 (56.1%)  
  
New Test (n=3453):  
Illegal (0): 1516 (43.9%)  
Legal (1): 1937 (56.1%)
```

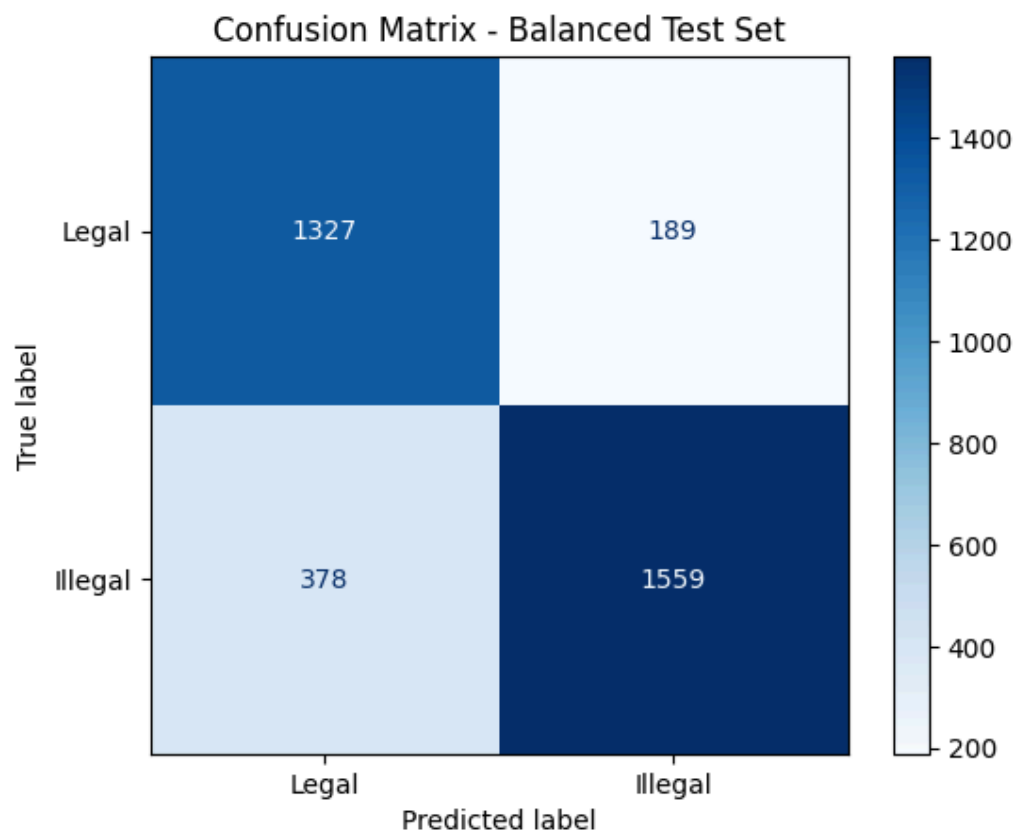
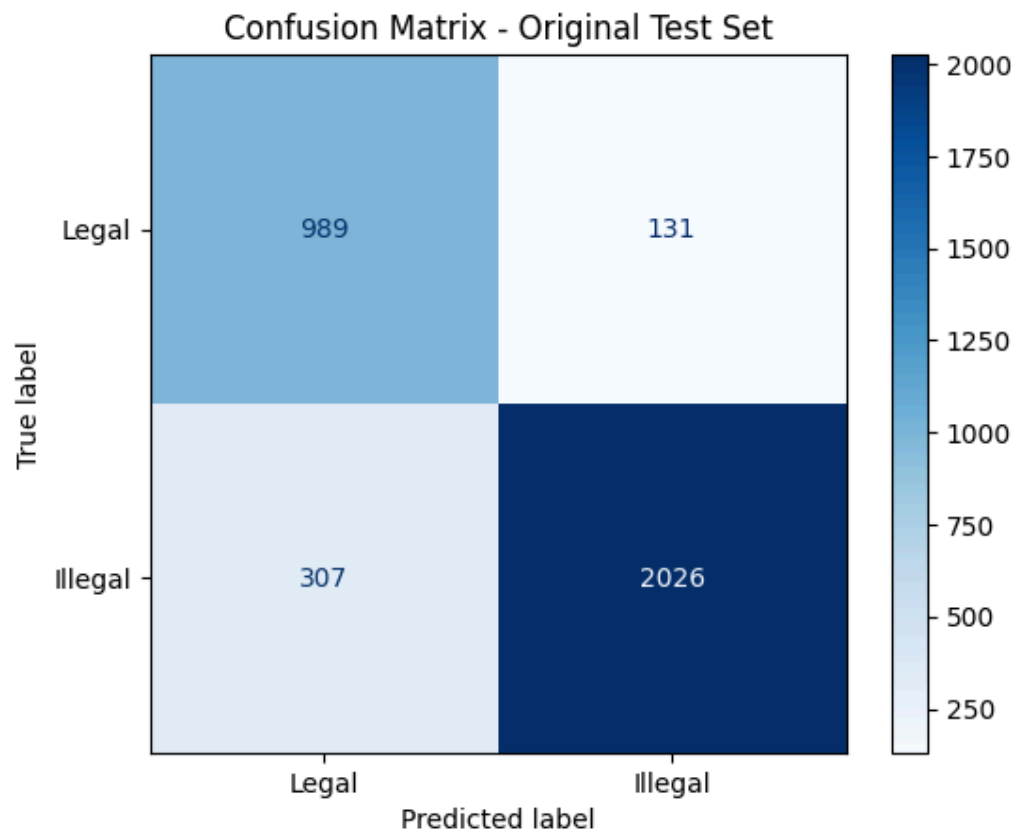
Results for Polish BERT without weights, trained on original datasets given by HuggingFace author of dataset.

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.269100	0.627923	0.712969	0.303514	0.558824	0.208333
2	0.171200	0.607225	0.743910	0.437048	0.642553	0.331140
3	0.110100	0.923573	0.747202	0.525926	0.601695	0.467105

```
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Original Test Set Results:
Accuracy: 0.8732
F1: 0.9024
Precision: 0.9393
Recall: 0.8684
```

```
Balanced Test Set Results:
Accuracy: 0.8358
F1: 0.8461
Precision: 0.8919
Recall: 0.8049
```

Confusion Matrix for Polish BERT:



Results for Base BERT without weights, trained on original datasets given by HuggingFace author of dataset.

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.273100	0.661926	0.639236	0.450902	0.415129	0.493421
2	0.263400	0.655593	0.713627	0.406548	0.537906	0.326754
3	0.214300	0.826599	0.714286	0.495349	0.527228	0.467105

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Original Test Set Results:

Accuracy: 0.8613

F1: 0.8937

Precision: 0.9268

Recall: 0.8628

Balanced Test Set Results:

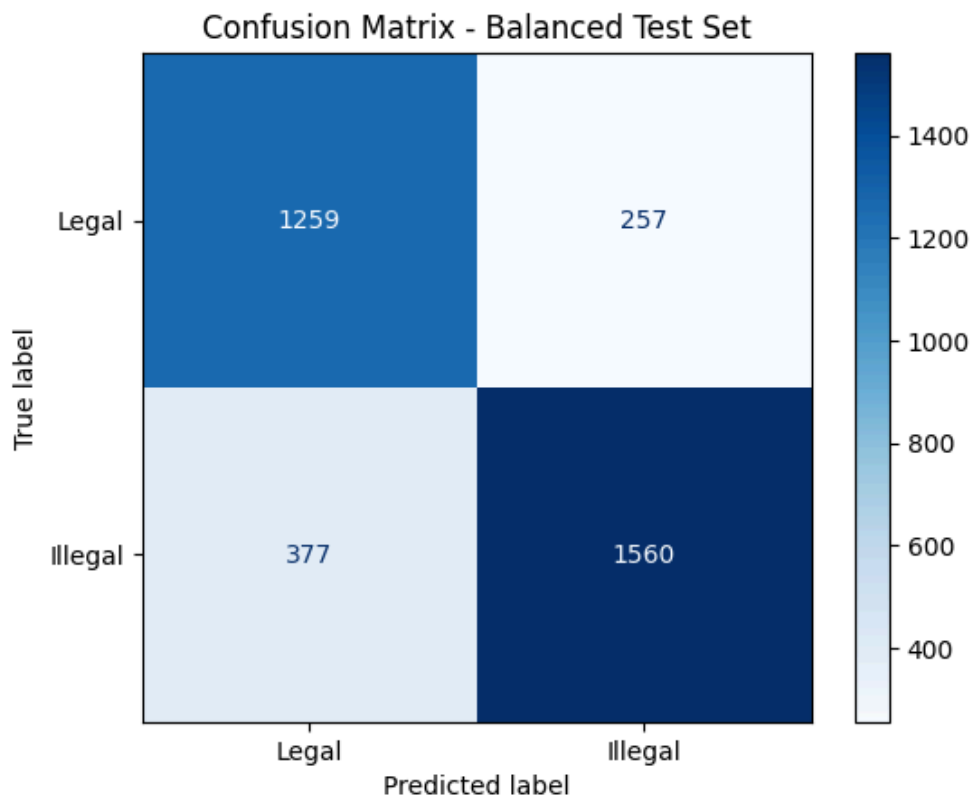
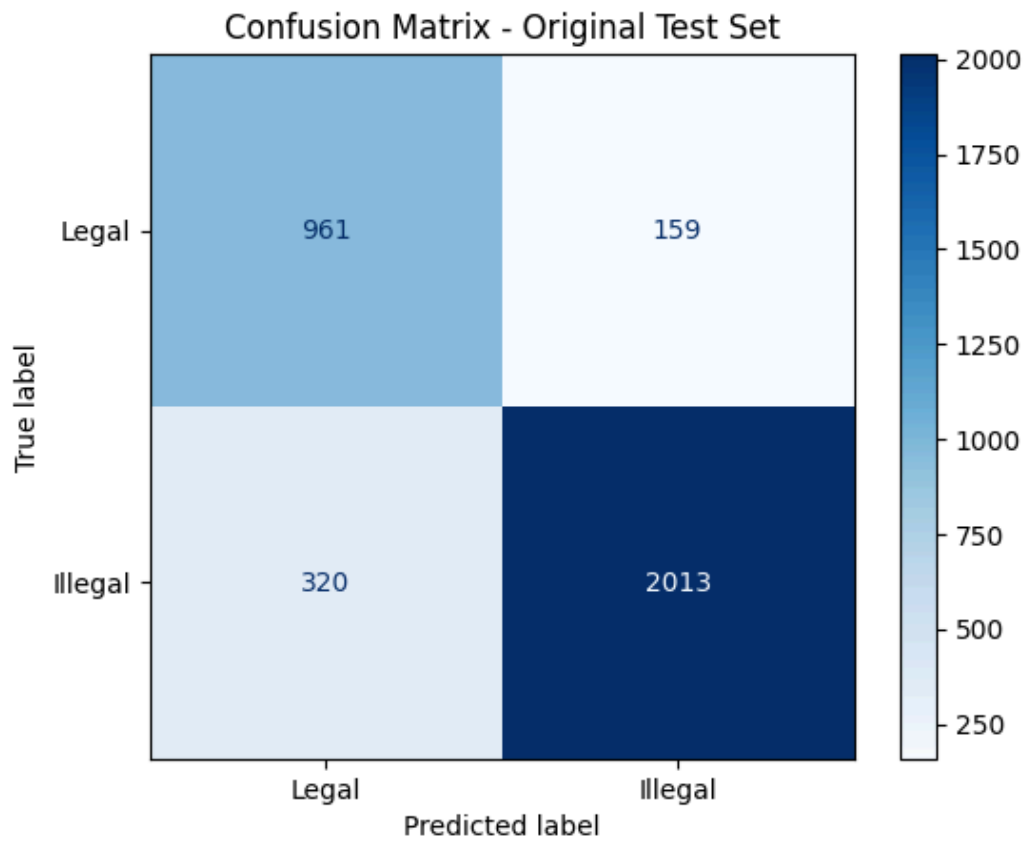
Accuracy: 0.8164

F1: 0.8311

Precision: 0.8586

Recall: 0.8054

Confusion Matrix for Base BERT:



Results for Legal BERT without weights, trained on original datasets given by HuggingFace author of dataset.

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	0.351100	0.728548	0.620803	0.304348	0.338710	0.276316
2	0.411700	0.697563	0.685978	0.149733	0.400000	0.092105
3	0.406600	0.656503	0.684661	0.243286	0.435028	0.168860

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Original Test Set Results:

Accuracy: 0.8147

F1: 0.8588

Precision: 0.8849

Recall: 0.8341

Balanced Test Set Results:

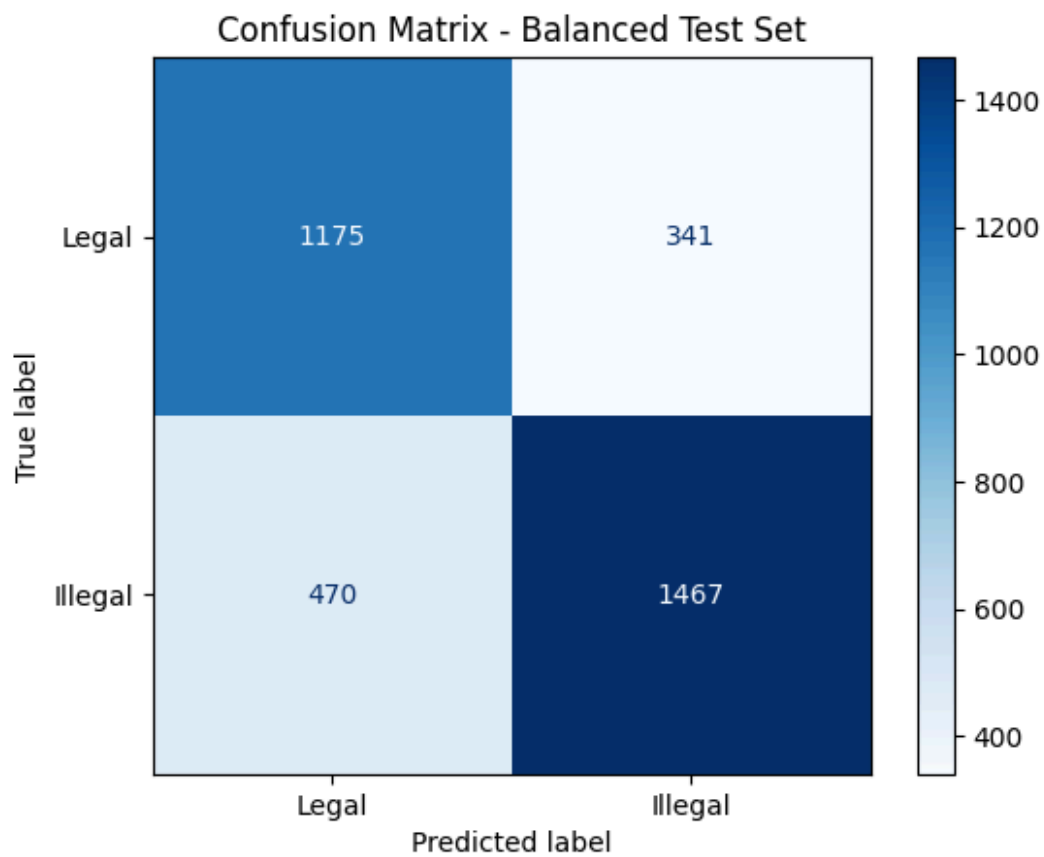
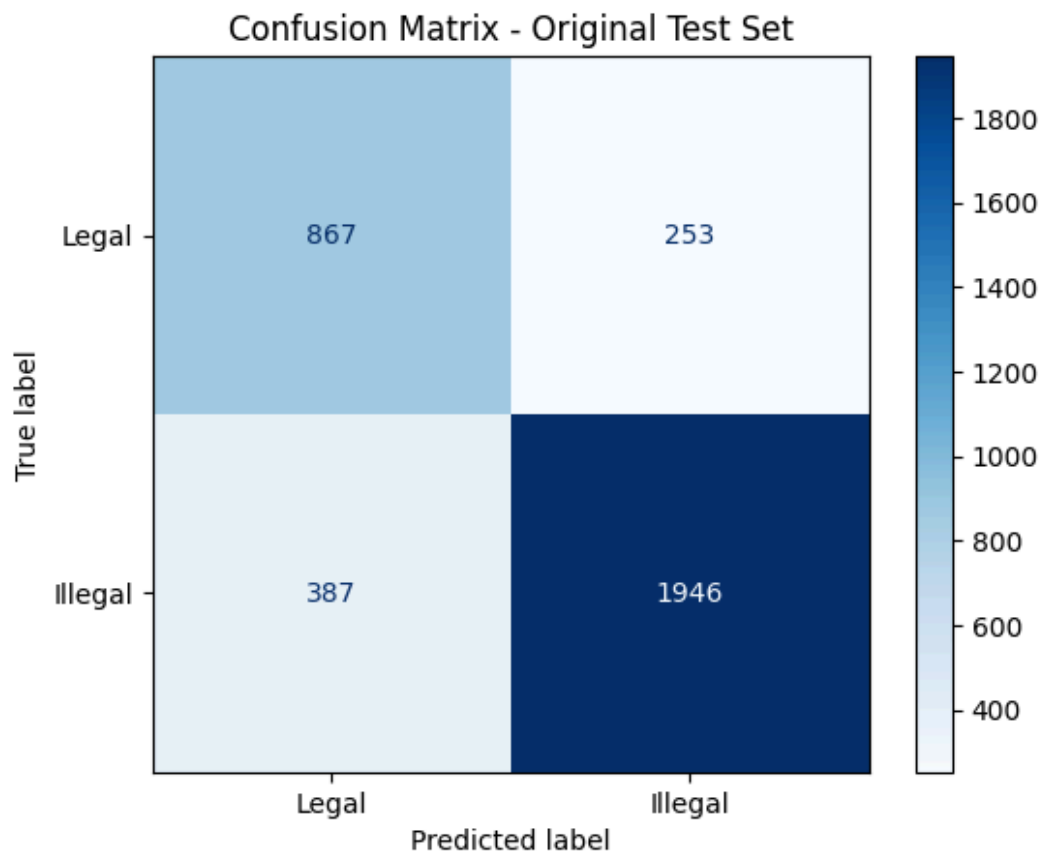
Accuracy: 0.7651

F1: 0.7834

Precision: 0.8114

Recall: 0.7574

Confusion Matrix for Base BERT:



For Polish BERT without adding weights with using balanced validation set:

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Balanced test Results:

Accuracy: 0.8271

F1: 0.8222

Precision: 0.7496

Recall: 0.9103

Confusion Matrix:

[[1380 136]

[461 1476]]

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Original test Results:

Accuracy: 0.8674

F1: 0.8183

Precision: 0.7364

Recall: 0.9205

Confusion Matrix:

[[1031 89]

[369 1964]]

For Polish BERT with adding weights with using balanced validation set:

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Balanced test Results:

Accuracy: 0.8288

F1: 0.8259

Precision: 0.7461

Recall: 0.9248

Confusion Matrix:

[[1402 114]

[477 1460]]

=====

Original test Results:

Accuracy: 0.8624

F1: 0.8134

Precision: 0.7263

Recall: 0.9241

Confusion Matrix:

[[1035 85]

[390 1943]]

After testing multiple approaches, the transformer-based Polish BERT without weights, trained on original datasets showed the best results, achieving an F1-score of around 0.90 on the original given test set and around 0.84 on my balanced test set.

My approach for balancing validation sets and giving weights did not improve the model by any means. I assume the validation set given by the author was made unbalanced on purpose, so the model will train better. Test set is unbalanced in favor of legal cases, probably because in real life most cases are legal, so it mimics real life scenarios.

We can clearly see that BERT fine-tuned for Polish words worked the best with Polish legal cases. It far surpassed the base and legal model. Legal model had the worst scores, maybe because the Polish legal structure is not similar to the English that this model was fine-tuned to.